

WEEK 7

STUDENT PERFORMANCE - DATA CLEANING AND TRANSFORMATION

Introduction :

The Student Performance dataset was prepared and transformed using Power BI and Power Query Editor. The data was cleaned, structured into separate tables, transformed using unpivoting, and conditional columns were created for student performance analysis. A data model was then created using relationships between the tables, followed by the creation of DAX measures.

Objectives :

- To clean and prepare the Student Performance dataset.
- To create a unique Student ID using an Index Column.
- To organize the data into separate tables.
- To transform subject-wise marks using unpivoting.
- To create conditional columns for Result and Grade.
- To establish relationships between the tables.
- To create DAX measures for performance analysis.

Dataset Preparation :

The Student Performance dataset was imported into Power BI and opened in Power Query Editor. The dataset was reviewed and prepared for further transformation. The required data was organized and structured to make it suitable for creating a data model.

Query Duplication and Student ID :

The original query was duplicated to create separate data structures for different purposes. An Index Column was created and used as the Student ID to uniquely identify each student.

The indexed data was further duplicated to create the required tables for the data model.

Creation of Tables :

Three main tables were created from the prepared dataset:

➤ Students Details

The Students Details table contains student-related information such as:

- Student ID
- Gender
- Race/Ethnicity
- Parental Level of Education
- Lunch
- Test Preparation Course
- Average
- Result
- Grade

Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course	average score
1	Female	Group B	Bachelor's Degree	Standard	No	
2	Female	Group C	Some College	Standard	Co	
3	Female	Group B	Master's Degree	Standard	No	
4	Male	Group A	Associate's Degree	Free/Reduced	No	
5	Male	Group C	Some College	Standard	No	
6	Female	Group B	Associate's Degree	Standard	No	
7	Female	Group B	Some College	Standard	Co	
8	Male	Group B	Some College	Free/Reduced	No	
9	Male	Group D	High School	Free/Reduced	Co	
10	Female	Group B	High School	Free/Reduced	No	
11	Male	Group C	Associate's Degree	Standard	No	
12	Male	Group D	Associate's Degree	Standard	No	
13	Female	Group B	High School	Standard	No	
14	Male	Group A	Some College	Standard	Co	
15	Female	Group A	Master's Degree	Standard	No	
16	Female	Group C	Some High School	Standard	No	
17	Male	Group C	High School	Standard	No	
18	Female	Group B	Some High School	Free/Reduced	No	
19	Male	Group C	Master's Degree	Free/Reduced	Co	
20	Female	Group C	Associate's Degree	Free/Reduced	No	

➤ StudentsPerformance

The StudentsPerformance table contains the main student performance information, including:

- Student ID
- Gender
- Race/Ethnicity
- Parental Level of Education
- Lunch
- Test Preparation Course
- Math Score
- Reading Score
- Writing Score
- Average

- Result
- Grade

Queries [4] fx Table.TransformColumns(#"Changed Type2",{{"Average", each Number.Round(., 2), type number}}) Query Settings

Student_ID	gender	race/ethnicity	parental level of education	lunch	te
1	Female	Group B	Bachelor's Degree	Standard	No
2	Female	Group C	Some College	Standard	Co
3	Female	Group B	Master's Degree	Standard	No
4	Male	Group A	Associate's Degree	Free/Reduced	No
5	Male	Group C	Some College	Standard	No
6	Female	Group B	Associate's Degree	Standard	No
7	Female	Group B	Some College	Standard	Co
8	Male	Group B	Some College	Free/Reduced	No
9	Male	Group D	High School	Free/Reduced	Co
10	Female	Group B	High School	Free/Reduced	No
11	Male	Group C	Associate's Degree	Standard	No
12	Male	Group D	Associate's Degree	Standard	No
13	Female	Group B	High School	Standard	No
14	Male	Group A	Some College	Standard	Co
15	Female	Group A	Master's Degree	Standard	No
16	Female	Group C	Some High School	Standard	No
17	Male	Group C	High School	Standard	No
18	Female	Group B	Some High School	Free/Reduced	No
19	Male	Group B	Master's Degree	Free/Reduced	Co
20	Female	Group C	Associate's Degree	Free/Reduced	No
21					

PROPERTIES
Name: StudentsPerformance

APPLIED STEPS
Source
Promoted Headers
Changed Type
Trimmed Text
Capitalized Each Word
Cleaned Text
Removed Duplicates
Added Index
Renamed Columns
Reordered Columns
Added Custom
Added Custom1
Changed Type1
Added Conditional Column
Added Conditional Column1
Changed Type2
Rounded Off

➤ Students Marks

The Students Marks table was created for subject-wise mark analysis. The subject scores were transformed into a structured format using unpivoting.

Queries [4] fx Table.RenameColumns(#"Unpivoted Columns",{{"Attribute", "Subjects"}, {"Value", "Score"}}) Query Settings

Student_ID	Subjects	Score
1	math score	72
2	reading score	72
3	writing score	74
4	math score	69
5	reading score	90
6	writing score	88
7	math score	90
8	reading score	95
9	writing score	93
10	math score	47
11	reading score	57
12	writing score	44
13	math score	76
14	reading score	78
15	writing score	75
16	math score	71
17	reading score	83
18	writing score	78
19	math score	88
20	reading score	95
21	writing score	92

PROPERTIES
Name: StudentsMarks

APPLIED STEPS
Promoted Headers
Changed Type
Trimmed Text
Capitalized Each Word
Cleaned Text
Removed Duplicates
Added Index
Renamed Columns
Reordered Columns
Added Custom
Added Custom1
Changed Type1
Added Conditional Column
Added Conditional Column1
Changed Type2
Rounded Off
Removed Other Columns
Unpivoted Columns
Renamed Columns1

Unpivoting Subject Scores :

The Math Score, Reading Score, and Writing Score columns were unpivoted in Power Query.

After unpivoting:

- The subject names were placed in the Subjects column.
- The corresponding marks were placed in the Score column.
- Student ID was retained to identify the student.

This transformation converted the subject-wise marks into a suitable format for further analysis.

Before Unprivoting

Student_ID	math score	reading score	writing score
1	72	72	74
2	69	90	88
3	90	95	93
4	47	57	44
5	76	78	75
6	71	83	78
7	88	95	92
8	40	43	39
9	64	64	67
10	38	60	50
11	58	54	52
12	40	52	43
13	65	81	73
14	78	72	70
15	50	53	58
16	69	75	78
17	88	89	86
18	18	32	28
19	46	42	46
20	54	58	61
21	66	69	63

After Unprivoting

Student_ID	Subjects	Score
1	1 math score	72
2	1 reading score	72
3	1 writing score	74
4	2 math score	69
5	2 reading score	90
6	2 writing score	88
7	3 math score	90
8	3 reading score	95
9	3 writing score	93
10	4 math score	47
11	4 reading score	57
12	4 writing score	44
13	5 math score	76
14	5 reading score	78
15	5 writing score	75
16	6 math score	71
17	6 reading score	83
18	6 writing score	78
19	7 math score	88
20	7 reading score	95
21	7 writing score	92

Conditional Columns :

Conditional columns were created to classify student performance.

➤ Result

A conditional column was created to identify whether a student passed or failed based on the Average score.

- If Average is greater than 40 → Pass
- Otherwise → Fail

Add Conditional Column

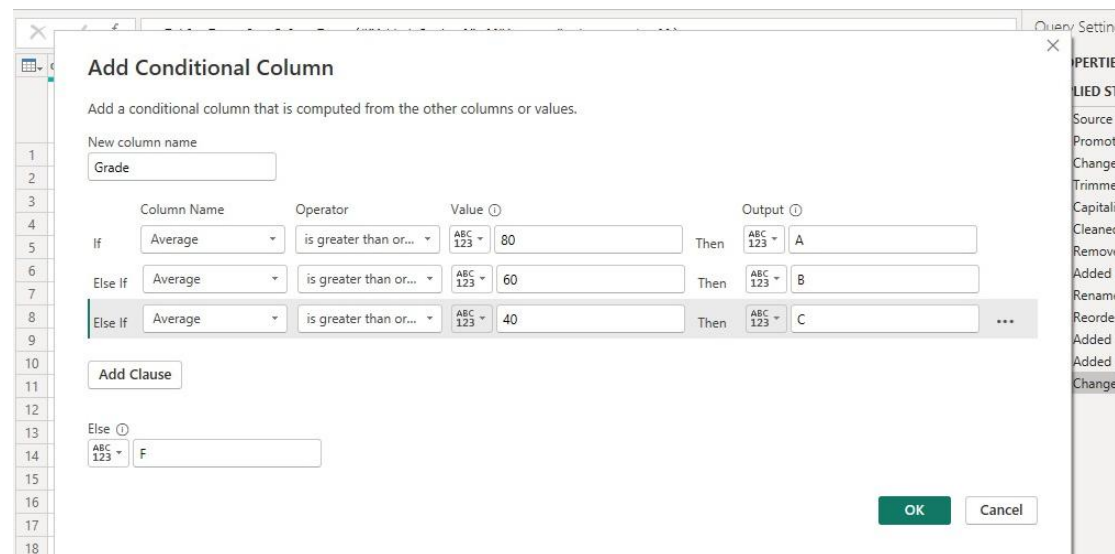
Add a conditional column that is computed from the other columns or values.

New column name:

Column Name	Operator	Value	Output
If	Average	is greater than or...	40
Then	ABC 123	Pass	
Else	ABC 123	Fail	

➤ Grade

A Grade conditional column was created using conditional logic based on the student's Average score. This column classifies students into different performance grades.



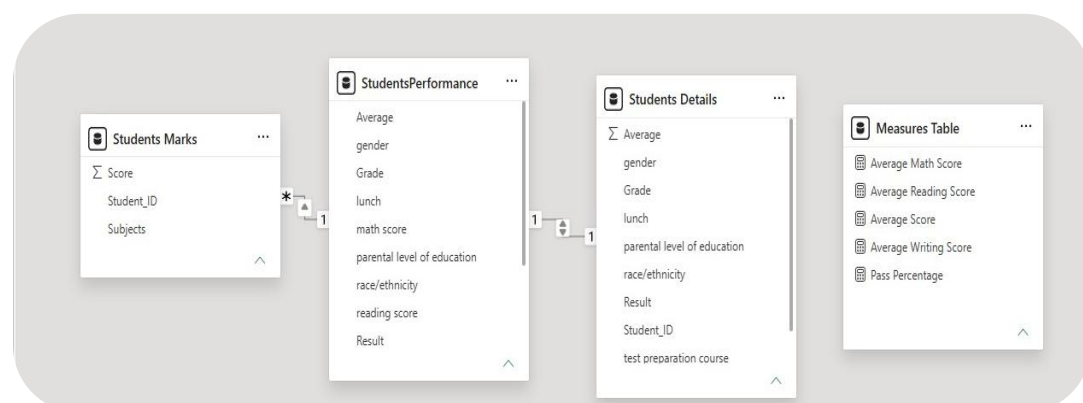
Data Modeling and Relationships :

The three tables were connected in the Model view using Student ID.

The relationship between Students Marks and StudentsPerformance was created so that the subject-wise marks could be connected with the main student performance data.

The relationship between StudentsPerformance and Students Details was also created to connect the student information with the performance information.

A separate Measures table was created to store all the DAX measures in one place. The Measures table does not require a relationship with the other tables.



DAX Measures :

A separate Measures table was created and the following DAX measures were added.

➤ Average Math Score

Average Math Score = `AVERAGE(StudentsPerformance[math score])`

➤ Average Reading Score

Average Reading Score = `AVERAGE(StudentsPerformance[reading score])`

➤ Average Score

Average Score = `AVERAGE('StudentsPerformance'[Average])`

➤ Average Writing Score

Average Writing Score = `AVERAGE(StudentsPerformance[writing score])`

➤ Pass Percentage

Pass Percentage =
`DIVIDE(CALCULATE(COUNT(StudentsPerformance[Result]),'StudentsPerformance'[Result]="Pass"),DISTINCTCOUNT('Students Details'[Student_ID]),0)`

Data Validation :

The transformed tables, calculated columns, relationships, and DAX measures were checked to ensure that the data model was properly structured and the calculated results were generated correctly.

Final Output :

The final Power BI data model consists of:

- Students Details table
- StudentsPerformance table
- Students Marks table
- Measures table
- Student ID for identifying students
- Unpivoted subject-wise marks
- Result and Grade conditional columns
- Relationships between the required tables
- Five DAX measures for performance analysis

The dataset is now cleaned, transformed, structured, and ready for the visualization and dashboard activities in Week 8.

Conclusion :

The Student Performance dataset was successfully cleaned and transformed using Power Query. Separate tables were created, subject scores were unpivoted, conditional columns were added, relationships were established, and DAX measures were created in a dedicated Measures table. The completed data model provides a structured foundation for the Week 8 visualization and dashboard activities.